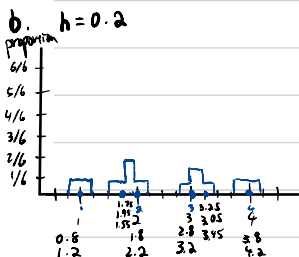
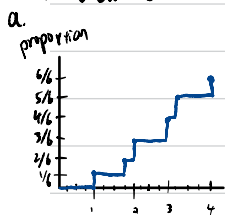
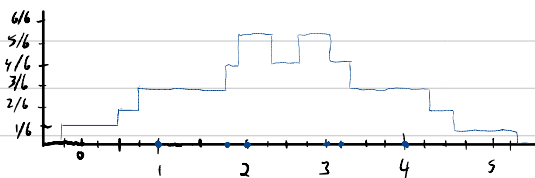


Homework: Kernel Density Estimation

Problem 1



d.



$h = 1.3$

-0.3
 2.3
 4.3
 5.3
 6.3
 7.3
 8.3
 9.3
 10.3
 11.3
 12.3
 13.3
 14.3
 15.3
 16.3
 17.3
 18.3
 19.3
 20.3
 21.3
 22.3
 23.3
 24.3
 25.3
 26.3
 27.3
 28.3
 29.3
 30.3
 31.3
 32.3
 33.3
 34.3
 35.3
 36.3
 37.3
 38.3
 39.3
 40.3
 41.3
 42.3
 43.3
 44.3
 45.3
 46.3
 47.3
 48.3
 49.3
 50.3

c. $h_0 = 1.84 \times \text{sd}(X) \times n^{-1/5}$

mean: $\frac{4.0 + 3.25 + 3.0 + 2.0 + 1.75 + 1.0}{6} = 2.5$

$$\frac{(4 - 2.5)^2 + (3.25 - 2.5)^2 + (3 - 2.5)^2 + (2 - 2.5)^2 + (1.75 - 2.5)^2 + (1 - 2.5)^2}{6} = 1.5$$

$$\frac{15}{6} = \frac{5}{2} = 2.5$$

$$\text{sd} \approx 1.01$$

$$1.84 \times 1.01 \times 6^{-1/5} \approx 1.299 \approx 1.3$$

e.

They are similar that since both are uniform they are block/box shaped. The KDE for part d covers values not covered in part b. Part

D's plot is one long "curve" while part b is split up into a few box/segments.

Problem 2

a.

$$\hat{f}_h(x) = \frac{\sum_{i=1}^n I\{|x_i - x| < h\}}{2nh} \rightarrow 2(5)(0.6)$$

$$\begin{array}{l} | -1 - 0 | < 0.6 \\ | -0.5 - 0 | < 0.6 \\ | 0.5 - 0 | < 0.6 \\ | 1 - 0 | < 0.6 \\ | 0 - 0 | < 0.6 \end{array} \left. \vphantom{\begin{array}{l} | -1 - 0 | < 0.6 \\ | -0.5 - 0 | < 0.6 \\ | 0.5 - 0 | < 0.6 \\ | 1 - 0 | < 0.6 \\ | 0 - 0 | < 0.6 \end{array}} \right\} \text{true} \quad \frac{3}{2(5)(0.6)} = \frac{3}{6} = \boxed{\frac{1}{2}}$$

b. $E[\hat{f}_h(0)]$ $h=0.6$ true

$$\frac{4}{5} \cdot \frac{1}{5} = \frac{3}{5} = 0.6$$

$$= F_x(0+0.6) - F_x(0-0.6)$$

$$\frac{F_x(0.6) - F_x(-0.6)}{2(0.6) = 1.2} = \frac{0.6}{1.2} = \boxed{\frac{1}{2}}$$

-1, -0.5, 0, 0.5, 1

$$c. f(x) = \frac{1}{6\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{6}\right)^2} = \frac{1}{0.707\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{0-0}{0.707}\right)^2} \approx 0.564$$

$$\mu = \begin{array}{l} -1.0 \\ -0.5 \\ 0.0 \\ 0.5 \\ +1 \end{array} \quad 6 = (-1)^2 + (-0.5)^2 + (0)^2 + (0.5)^2 + (1)^2$$

$$1 + 0.25 + 0.25 + 1 = 2.5$$

$$\sqrt{\frac{2.5}{5}} \approx 0.707$$

$$\frac{+1}{0/5 = 0}$$

d. No, they are not equal. This indicates that uniform KDE ^{may} be biased as $E[\hat{f}_h(0)] \neq f_h(0)$

e. $E[\hat{f}_h(0)]$ $-1, -0.5, 0, 0.5, 1$

The gap is growing because there is only one data point that falls within ± 0.1 , and it is overfitting.

$$\frac{F_x(0.1) - F_x(-0.1)}{2(0.1)} = \frac{3/5 - 2/5}{0.2} = \frac{1/5}{0.2} = 0.2 = \boxed{1}$$

f. Yes, $\hat{f}_h$ is a consistent estimator because

as n grows large, and h goes to 0, it approaches the true (population) value.

g. You want to make sure h is not too large where it's underfitting the data, but not too small that it leads to the issue of overfitting. Those are the main considerations.

Problem 3 $X = [5, 7, 7, 8, 10]$

a. $h = 1.5$

$$\hat{f}_h(6) = \frac{3}{2(5)(1.5)} = \frac{3}{15} = \boxed{\frac{1}{5}}$$

$$\hat{f}_h(x) = \frac{\sum_{i=1}^n \mathbb{I}\{|x_i - x| < h\}}{2nh} \rightarrow \frac{1}{2(5)(0.6)}$$

$$|5 - 6| = 1$$

$$|7 - 6| = 1$$

$$|7 - 6| = 1$$

$$|8 - 6| = 2$$

$$|10 - 6| = 4$$

$$\hat{f}_h(6.5) = \frac{2}{15}$$

$$|5 - 6.5| = 1.5$$

$$|7 - 6.5| = 0.5$$

$$|7 - 6.5| = 0.5$$

$$|8 - 6.5| = 1.5$$

$$|10 - 6.5| = 3.5$$

$$|5 - 8| = 3$$

$$|7 - 8| = 1$$

$$|7 - 8| = 1$$

$$|8 - 8| = 0$$

$$|10 - 8| = 2$$

$$\hat{f}_h(7) = \frac{3}{15} = \boxed{\frac{1}{5}}$$

$$|5 - 7| = 2$$

$$|7 - 7| = 0$$

$$|7 - 7| = 0$$

$$|8 - 7| = 1$$

$$|10 - 7| = 3$$

$$\hat{f}_h(8) = \frac{3}{15} = \boxed{\frac{1}{5}}$$

$$|5 - 8.5| = -3.5$$

$$|7 - 8.5| = 1.5$$

$$|7 - 8.5| = 1.5$$

$$|8 - 8.5| = 0$$

$$|10 - 8.5| = 1.5$$

$$\hat{f}_h(8.5) = \frac{1}{15}$$

$$x = [5, 7, 7, 8, 10]$$

$$h = 1.5$$

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x_i - x}{h} \right)^2}$$

$$y = 5 - 6$$

$$y = -1$$

$$y = 7 - 6 = 1$$

$$y = 7 - 6 = 1$$

$$y = 8 - 6 = 2$$

$$y = 10 - 6 = 4$$

$$\hat{f}_h(6) = \frac{1}{5(1.5)}$$

$$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{5-6}{1.5} \right)^2}$$

$$\frac{1}{5(1.5)} \left[\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{5-6}{1.5} \right)^2} + \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{7-6}{1.5} \right)^2} + \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{7-6}{1.5} \right)^2} + \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{8-6}{1.5} \right)^2} + \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{10-6}{1.5} \right)^2} \right] = 0.151$$

$$\hat{f}_h(6.5) =$$

$$\begin{aligned} 5 - 6.5 &= -1.5 \\ 7 - 6.5 &= 0.5 \\ 7 - 6.5 &= 0.5 \\ 8 - 6.5 &= 1.5 \\ 10 - 6.5 &= 3.5 \end{aligned} \quad \frac{1}{5(1.5)} \left[\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{-1.5}{1.5} \right)^2} + \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{0.5}{1.5} \right)^2} + \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{0.5}{1.5} \right)^2} + \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{1.5}{1.5} \right)^2} + \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{3.5}{1.5} \right)^2} \right]$$

$$\hat{f}_h(7) = \left[\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{1.5}{1.5} \right)^2} + \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{3.5}{1.5} \right)^2} \right] = 0.169$$

$$\begin{aligned} 5 - 7 &= -2 \\ 7 - 7 &= 0 \\ 7 - 7 &= 0 \end{aligned} \quad \frac{1}{5(1.5)} \left[\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{-2}{1.5} \right)^2} + \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{0}{1.5} \right)^2} + \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{0}{1.5} \right)^2} + \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{6}{1.5} \right)^2} + \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{6}{1.5} \right)^2} \right]$$

$$\begin{aligned} 8 - 7 &= 1 \\ 10 - 7 &= 3 \end{aligned} \quad \left[\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{1}{1.5} \right)^2} + \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{3}{1.5} \right)^2} \right] = 0.178$$

$$\hat{f}_h(8) = \frac{1}{5(1.5)} \left[\frac{1}{\sqrt{2\pi}} e^{\left(-\frac{1}{2}\left(\frac{3}{1.5}\right)^2\right)} + \frac{1}{\sqrt{2\pi}} e^{\left(-\frac{1}{2}\left(\frac{-1}{1.5}\right)^2\right)} + \frac{1}{\sqrt{2\pi}} e^{\left(-\frac{1}{2}\left(\frac{-1}{1.5}\right)^2\right)} + \frac{1}{\sqrt{2\pi}} e^{\left(-\frac{1}{2}\left(\frac{0}{1.5}\right)^2\right)} + \frac{1}{\sqrt{2\pi}} e^{\left(-\frac{1}{2}\left(\frac{3}{1.5}\right)^2\right)} \right] = 0.167$$

$5 - 8 = -3$
 $7 - 8 = -1$
 $7 - 8 = -1$
 $8 - 8 = 0$
 $10 - 8 = 2$

$$\hat{f}_h(8.5) = \frac{1}{5(1.5)} \left[\frac{1}{\sqrt{2\pi}} e^{\left(-\frac{1}{2}\left(\frac{-3.5}{1.5}\right)^2\right)} + \frac{1}{\sqrt{2\pi}} e^{\left(-\frac{1}{2}\left(\frac{-1.5}{1.5}\right)^2\right)} + \frac{1}{\sqrt{2\pi}} e^{\left(-\frac{1}{2}\left(\frac{-1.5}{1.5}\right)^2\right)} + \frac{1}{\sqrt{2\pi}} e^{\left(-\frac{1}{2}\left(\frac{0.5}{1.5}\right)^2\right)} + \frac{1}{\sqrt{2\pi}} e^{\left(-\frac{1}{2}\left(\frac{1.5}{1.5}\right)^2\right)} \right] = 0.151$$

$5 - 8.5 = -3.5$
 $7 - 8.5 = -1.5$
 $7 - 8.5 = -1.5$
 $8 - 8.5 = 0.5$
 $10 - 8.5 = 1.5$

c.

Part a	$\frac{2}{15}$	Part b
$\hat{f}_h(6) = \frac{1}{15}$	$\frac{1}{15}$	$\hat{f}_h(6) = 0.151$
$\hat{f}_h(6.5) = \frac{2}{15}$	$\frac{1}{15}$	$\hat{f}_h(6.5) = 0.167$
$\hat{f}_h(7) = \frac{1}{15}$	$\frac{2}{15}$	$\hat{f}_h(7) = 0.178$
$\hat{f}_h(8) = \frac{1}{15}$	$\frac{2}{15}$	$\hat{f}_h(8) = 0.167$
$\hat{f}_h(8.5) = \frac{1}{15}$	$\frac{2}{15}$	$\hat{f}_h(8.5) = 0.151$

Biggest jump: $\frac{2}{15}$

The biggest jump in part a happens between $\hat{f}_h(8)$ and $\hat{f}_h(8.5)$. The shape of the Gaussian comes from the fact that each data point is turned into its normal distribution and each x point is "spread" out to create a smooth curve, as the normal distribution overlaps. The jump exists in the uniform because in the uniform the data point falls within the bandwidth or not - there is no spread of data.

d. You might prefer the Gaussian if you want to guess how the distribution would have been like if we had more data - guessing using the normal distribution. You might prefer uniform if the data is sparse as you don't want to misrepresent the data by having some density in places where there weren't any originally by spreading the data.